MILITARY SPECIFICATION

HOSE ASSEMBLIES, HYDRAULIC, NONMAGNETIC,

FOR MINESWEEPING TENSIOMETERS

1 SCOPE

1.1 Scope. - This specification covers a nonmagnetic hydraulic hose assembly used with remote reading minesweeping tensiometers.

2 APPLICABLE DOCUMENTS

2.1 Other publications. - The following document forms a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids or request for proposal shall apply.

OFFICIAL CLASSIFICATION COMMITTEE

Uniform Freight Classification Ratings, Rules and Regulations.

(Application for copies should be addressed to the Official Classification Committee, 1 Park Avenue at 33rd Street, New York, N. Y. 10016.)

3 REQUIREMENTS

3.1 General requirements. -

3.1.1 Hose assembly. - The hose assembly shall consist of the following:

(a) A hydraulic hose which incorporates an inner tube, a reinforcement and a cover.
(b) Two hose coupling fittings, one installed at each end of the hose for connecting the hose to the tensiometer gage and cylinder.
(c) Two helical springs, one installed on each hose coupling, to prevent excessive bending of the hose at the fitting.

3.1.2 Screw and pipe threads. - All screw and pipe threads shall be commercial standard (see 3.2.4).

3.1.3 Operating pressure. - The hose assembly shall be capable of operation in a 3000 pounds per square inch (psi) hydraulic system.

3.1.4 Permeability. - The hose braid, end fittings, and springs of the hose assembly shall be fabricated from austenitic steel of low magnetic permeability.

3.2 Construction. -

3.2.1 Inner tube. - The inner tube shall be oil resistant, synthetic rubber of seamless construction and uniform gage. It shall have a smooth bore, free from pitting, dirt, foreign material, or mandrel lubricants as is consistent with good manufacturing practice.

3.2.2 Reinforcement. - The reinforcement shall include an inner cotton braid and a wire braid, braided under proper tension, in order to limit the cubical expansion of the hose to a minimum, and to provide sufficient strength to meet the other requirements specified herein. The wire braid shall be of nonmagnetic austenitic corrosion resisting steel. No length of hose shall contain broken or spliced reinforcing wires, and no wires shall be omitted from the braid.

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3.2.3 Cover. - The cover shall be abrasion, oil and mildew resistant synthetic rubber. A friction ply or braid may be used as a backing for the rubber.

3.2.4 End fittings. - The hose coupling end fittings shall be reuseable male pipe type consisting of a shell and nipple. The shell shall be provided with a coarse internal thread, so configured that it may be threaded on to the hose, and a 1/2-inch, 20 NF-2 internal thread for assembly with the nipple. The outside of the shell shall be hexagonal in shape, about 5/8-inch across flats, and shall be grooved at the hose end for the hose spring. The hose end of the nipple shall be designed to permit insertion of the nipple in the hose and shall have a 1/2-inch, 20 NF2 external threaded section for assembly in the shell. The opposite end of the nipple shall have a male 1/4-inch, 18 National Pipe Thread for connecting the hose assembly to the tensiometer components and a hexagonal section to facilitate assembly of the nipple with the hose and shell. Permeability shall be as specified in 3.1.4.

3.2.5 Springs. - The springs shall be of helical design and shall fit snugly on the hose. Permeability shall be as specified in 3.1.4.

3.3 Dimensions. -

3.3.1 Concentricity. - The outer surface of the cover shall be concentric with the inside bore of the hose within 0.030 inch total indicator reading. The variation in concentricity between the inside diameter (i.d.) of the inner tube and the outside diameter (o.d.) of the reinforcement shall not exceed this value.

3.3.2 Diameter. - The i.d. of the hose shall be 0.250 inches plus 0.023, minus 0.008 inches. The o.d. of the hose shall be 0.688 inches plus or minus 0.031 inches.

3.3.3 Lengths. - The lengths and tensiometer capacities shall be as shown in table I (see 6.1). The spring shall be 2-3/4 to 3 inches long. The hose coupling end fitting shall not exceed 2-1/2 inches in length.

Table I - Hose assembly lengths and tensiometer capacities.

Lengths	Tensiometer capacities
(feet)	(pounds)
12	0 to 5,000
30	0 to 10,000
50	0 to 20,000

3.3.4 Bending radius. - The minimum bending radius of the hose at inside of bend shall be 3 inches to permit coiling the hose assembly in the tensiometer carrying case.

3.4 Performance. -

3.4.1 Working pressure. - The hose assembly shall be designed to operate in the hydraulic system of the tensiometer at a maximum working pressure of 3,000 psi.

3.4.2 Temperature range. - The hose assembly shall have satisfactory operating characteristics through a temperature range of 0°F. to 150°F. There shall be no cracking, crazing or appreciable deformation when tested in accordance with 4.2.4 and 4.2.5.

3.4.3 Proof pressure. - The hose assembly shall withstand a hydraulic proof pressure test of 6000 psi without evidence of failure or leakage from the hose or end fittings when tested in accordance with 4.2.6.

3.4.4 Burst pressure. - The hose assembly shall withstand hydraulic pressures up to 9,000 psi without bursting when tested in accordance with 4.2.7.

3.5 Age. - Bulk hose over one year old shall not be used in the fabrication of the hose assemblies.

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3.6 Identification. - Markings shall be gasoline, oil, and water-resistant. Each hose assembly shall be marked on the cover at three places, one near each end fitting and one at the center, with the following:

(a) Federal stock number as specified (see 6.1).
(b) Manufacturer's name.
(c) Year of manufacture.

3.7 Workmanship. - The hose assembly shall be free from all defects which would affect its proper functioning in service.

4 QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. - Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified, the supplier may utilize his own facilities or any commercial laboratory acceptable to the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.2 Quality conformance inspection. -

4.2.1 Initial inspection. - Prior to the sampling and inspection specified in 4.3.2 and 4.3.3, two hose assemblies shall be subjected to the examination and tests specified in 4.2.2 through 4.2.7.

4.2.2 Construction. - All components of the hose assemblies shall be examined to determine conformance with 3.2.

4.2.3 Dimensions. - The hoses and the springs and hose couplings shall be examined to verify conformance with 3.3. When determining the size of the hose, measurements shall be made at least 1/2 inch from the end of the hose and the hose shall be straight. The hose shall show no appreciable flattening when bent to the minimum radius specified in 3.3.4.

4.2.4 Low temperature range test. - The hose assemblies shall be filled with hydraulic fluid and placed in a cold chamber, the temperature of which shall be approximately 0°F., and allowed to remain at this temperature for 24 hours. Immediately after removal from the cold chamber, a portion of the hose shall be straightened out and flexed through 180 degrees to a bending radius of three inches at the inside of the bend. The time required for flexing the hose through the 180 degree bend shall be not more than four seconds. After returning to room temperature, the hose assemblies shall be subjected to the proof pressure test specified in 4.2.6, except that they shall not be blown out after testing.

4.2.5 High temperature range test. - After the low temperature test has been conducted (see 4.2.4), the hose assemblies shall be subjected to oven heat of 150°F. for a period of 4 hours. Immediately after removal from the oven, a portion of each hose shall be straightened out and flexed to the bending radius specified in 4.2.4. After returning to room temperature, the hose assemblies shall be subjected to the proof pressure test specified in 4.2.6 and the testing fluid blown out.

4.2.6 Proof pressure. - The hose assembly shall be subjected to a hydraulic pressure of 6000 psi. The test shall be made either with water, or with hydraulic fluid. Pressure shall be held not less than 30 seconds. All testing fluid shall be blown out from the hose assembly after testing.

4.2.7 Burst pressure test. - After the examination of 4.2.2 and 4.2.3 and the tests of 4.2.4 through 4.2.6, the hose assemblies shall be subjected to a burst pressure test. The burst pressure test shall be made with hydraulic fluid. The pressure shall be increased at a uniform rate of not less than 10,000 psi, nor more than 20,000 psi per minute until failure occurs. The test pressure at which failure occurs shall be recorded.

4.3 Sampling. -

4.3.1 Lot. - All hose assemblies of the same length offered for delivery at one time shall be considered a lot.

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4.3.2 Sampling for production inspection. - Sample hose assemblies shall be selected at random from each lot in accordance with table I for the production inspection of 4.3.3. Failure of any hose assembly to meet inspection requirements shall be cause for rejection of the lot represented by the sample.

Table I. - Sampling for production inspection.

Number of hose assemblies in lot	Number of hose assemblies in sample
2 to 10	1
11 to 50	2
51 to 100	3
101 to 250	6
251 to 500	10

4.3.3 Production inspection. - Each of the samples selected in accordance with 4.3.2 shall be visually examined for proper workmanship, assembly and external dimensions and shall be subjected to the proof pressure test specified in 4.2.6.

5 PREPARATION FOR DELIVERY

5.1 Domestic shipment and early equipment installation. -

5.1.1 Hose assemblies. -

5.1.1.1 Preservation and packaging. - Preservation and packaging which may be the contractor's commercial practice, shall be sufficient to afford adequate protection against deterioration and physical damage during shipment from the supply source to the using activity and until early installation. Each hose assembly shall be coiled separately in an interior package.

5.1.1.2 Packing. - Packing shall be accomplished in a manner which will insure acceptance by common carrier, at the lowest rate, and will afford protection against physical damage during direct shipment from the supply source to the using activity for early installation. The shipping containers or method of packing shall conform to the Uniform Freight Classification Ratings, Rules and Regulations or other carrier regulations as applicable to the mode of transportation and may conform to the contractor's commercial practice.

5.1.1.3 Marking. -

5.1.1.3.1 Interior packages. - The marking information on the interior packages shall be in accordance with the contractor's commercial practice. It shall include nomenclature, hose length and tensiometer capacity, Federal stock number, date of manufacture, contract or order number, and contractor's name.

5.1.1.3.2 Exterior shipping containers. - Exterior shipping containers shall be marked in accordance with the contractor's commercial practice and shall include nomenclature(s), Federal stock number(s), contract or order number, contractor's name and shipping destination.

5.2 Domestic shipment and storage or overseas shipment. - The requirements, and levels of preservation, packaging, packing and marking for shipment shall be specified by the procuring activity (see 6.1).

(5.2.1 The following provides various levels for protection during domestic shipment and storage or overseas shipment, which may be required when procurement is made.

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5.2.1.1 Preservation and packaging. -

5.2.1.1.1 Level A. - Preservation and packaging shall be in accordance with the following:

(a) Each coiled hose assembly shall be preserved against moisture by inserting it in a vapor waterproof plastic bag fabricated from either cellulose acetate, cellulose acetate butyrate, or cellulose propionate material at least 8 mils thick after drawing. The air volume in the bag shall be kept to a practical minimum when sealing.
(b) All hose assemblies regardless of length shall be coiled and packaged separately as follows:
(1) The inside diameter of the coil shall be not less than 6 inches.
(2) To prevent unwinding, the coil shall be taped at four equally spaced locations with three turns of plastic friction tape at least 1/2-inch wide.
(3) The coiled hose shall fit snuggly in the plastic bag.
(4) The coiled hose and bag shall be inserted in an interior fiberboard box, style RSC, type CF, weather-resistant class, variety SW, grade W6c, in accordance with PPP-B-636. The dimensions of the bos shall be such to accommodate the bagged coil without internal motion. The box shall be waterproofed in accordance with the appendix of PPP-B-636.

5.2.1.1.2 Level C. - The hose assemblies shall be preserved and packaged in accordance with the manufacturer's commercial practice.

5.2.2 Packing. - The hose assemblies shall be packed level A or B as specified in the contract or order. When practicable, shipping containers shall contain an identical number of hose assemblies of the same length, and shall be uniform in size. In addition, all containers shall be new and able to withstand storage, rehandling and reshipment without the necessity of repacking.

5.2.2.1 Level A. - The interior packages containing the hose assemblies shall be packed in nailed wooden boxes conforming to class 2, style 2 of PPP-B-621. Corrugated fiberboard liners shall be used with the interior packages, if required, to obtain a tight pack in the box. Closure of the box shall be with screws in lieu of nails and the box shall be strapped in accordance with the appendix to PPP-B-621. The gross weight of the box shall not exceed 125 pounds.

5.2.2.2 Level B. - The interior packages containing the hose assemblies shall be packed in nailed wooden boxes conforming to class 1, style 1, of PPP-B-621. The unit packages shall be tightly packed in the shipping container. Closure of the box shall be with screws in lieu of nails and the box shall be strapped in accordance with the appendix to the box specification. The gross weight of the box shall not exceed 200 pounds.

5.2.3 Marking. - Unless otherwise specified in the contract or order, marking shall be as specified in 5.2.3.1 and 5.2.3.2. The method and size of the markings shall be in accordance with MIL-STD-129.

5.2.3.1 Interior packages. - The marking information on the interior packages shall be in accordance with the contractor's commercial practice. It shall include nomenclature, hose length and tensiometer capacity, Federal stock number, date of manufacture, contract or order number, and contractor's name.

5.2.3.2 Exterior shipping containers. - Exterior shipping containers shall be marked in accordance with the contractor's commercial practice and shall include nomenclature(s), Federal stock number(s), contract or order number, contractor's name and shipping destination.

6 NOTES

6.1 Ordering data. - Procurement documents should specify the following:

(a) Title, number and date of this specification.
(b) Length of hose assembly and capacity of tensiometer (see 3.3.3).
(c) Federal stock number assigned to hose assembly (see 3.6).
(d) Preservation, packaging, packing and marking requirements, if other than those required by 5.1 (see 5.2).

Preparing activity:
Navy - SH
(Project 1075-N103SH)

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SPECIFICATION ANALYSIS SHEET

Form Approved
Budget Bureau No. 119-R004

INSTRUCTIONS

This sheet is to be filled out by personnel either Government or contractor, involved in the use of the specification in procurement of products for ultimate use by the Department of Defense. This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost. Comments and the return of this form will be appreciated. Fold on lines on reverse side, staple in corner, and send to preparing activity *(as indicated on reverse hereof)*.

SPECIFICATION

ORGANIZATION *(Of submitter)*

CITY AND STATE

CONTRACT NO.

QUANTITY OF ITEMS PROCURED

DOLLAR AMOUNT
$

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES.

2. COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3. IS THE SPECIFICATION RESTRICTIVE?

☐ YES ☐ NO IF "YES", IN WHAT WAY?

4. REMARKS *(Attach any pertinent data which may be of use in improving this specification. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity)*

SUBMITTED BY *(Printed or typed name and activity)*

DATE

DD FORM 1 APR 63 1426 REPLACES NAVSHIPS FORM 4863, WHICH IS OBSOLETE D14000

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